

Spring 2008, Math706 – Numerical Linear Algebra

Homework 3

1. Suppose $\mathbf{A}$ is a 202×202 matrix with $\|\mathbf{A}\|_2 = 100$ and $\|\mathbf{A}\|_F = 101$. Give the sharpest possible lower bound on $\kappa_2(\mathbf{A})$.
2. (a) Between an adjacent pair of nonzero IEEE single precision real numbers, how many IEEE double precision number are there?
 (b) The floating point system $\mathbf{F}$ includes many integers but not all of them. Give an exact formula for the smallest positive integer n that does not belong to $\mathbf{F}$. In particular, what are the values of n for IEEE single and double precision arithmetic?
3. Each of the following problems describes an algorithm implemented on a computer satisfying the axioms of floating point arithmetic. For each one, state whether the algorithm is *backward stable*, *stable but not backward stable*, or *unstable*, and prove it or at least give a reasonably convincing argument.
 - (a) Data: $x \in \mathbb{C}$. Solution: $2x$, computed as $x \oplus x$.
 - (b) Data: $x \in \mathbb{C}$. Solution: x^2 , computed as $x \otimes x$.
 - (c) Data: $x \in \mathbb{C} - \{0\}$. Solution: 1 , computed as $x \oslash x$.
 - (d) Data: $x \in \mathbb{C}$. Solution: 0 , computed as $x \ominus x$.
 - (e) Data: none. Solution: e , computed by summing $\sum_{k=0}^1 1/k!$ from left to right using $\otimes$ and $\oplus$, stopping when a summand is reached of magnitude $< \epsilon_{\text{machine}}$.
 - (f) Data: none. Solution: e , computed by the same algorithm as above except with the series summed from right to left.
 - (g) Data: none. Solution: π , computed by doing an exhaustive search to find the smallest floating point number x in the interval $[3, 4]$ such that $s(x) \otimes s(x') \leq 0$. Here $s(x)$ is the algorithm that calculate $\sin(x)$ stably in the given interval, and x' denotes the next floating point number after x in the floating point system.
4. (a) Let unitary matrices $\mathbf{Q}_1, \dots, \mathbf{Q}_k \in \mathbb{C}^{m \times m}$ be fixed and consider the problem of computing, for $\mathbf{A} \in \mathbb{C}^{m \times m}$, the product $\mathbf{B} = \mathbf{Q}_k \cdots \mathbf{Q}_1 \mathbf{A}$. Let the computation be carried out from right to left by straightforward operations on a computer. Show that this algorithm is backward stable.
 (b) Give an example to show that this result no longer holds if the unitary matrices $\mathbf{Q}_j$ are replaced by arbitrary matrices $\mathbf{X}_j \in \mathbb{C}^{m \times m}$.
5. Consider the Theorem 17.1 in Page 122. For any particular choice of norm $\|\cdot\|$, the bound (17.5) implies a more quantitative normwise bound than (17.4). Derive such bounds for the norms $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_\infty$.

6. Let

$$\mathbf{L} = \begin{bmatrix} 1 & & & & \\ -l_{2,1} & 1 & & & \\ -l_{3,1} & -l_{3,2} & 1 & & \\ \vdots & \vdots & \vdots & \ddots & \\ -l_{m,1} & -l_{m,2} & -l_{m,3} & \dots & 1 \end{bmatrix}$$

and define $\mathbf{M} = \mathbf{L}^{-1}$. Derive a formula for m_{ij} . Which entries of $\mathbf{L}$ does m_{ij} depend on?

7. The polynomial $p(x) = x^5 - 2x^4 - 3x^3 + 3x^2 - 2x - 1$ has three real zeros. Applying Newton's method to p with initial guess $x_0 = 0$ produces a series of estimates $x_1, x_2, x_3, \dots$ that convergence rapidly to a root $x_* \approx -0.315$.

- (a) Compute $x_1, \dots, x_6$ in floating point arithmetic with $\epsilon_{\text{machine}} \approx 10^{-16}$. How many digits do you estimate are correct in each of these numbers?
- (b) Compute $x_1, \dots, x_6$ again exactly with the aid of a symbolic algebra system in MATLAB. Each x_j is a rational number. How many digits are there in the numerator and the denominator for each j ?